

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Objective

Develop a dynamic pricing system that learns the best product price using **Thompson Sampling**. The system updates pricing decisions based on customer conversion rates and adjusts prices when competitors change their prices.

Step 1: Create Database

```
CREATE DATABASE PricingEngine;  
  
USE PricingEngine;
```

Step 2: Create Context Table

```
Create table Context (  
    ContextID INT PRIMARY KEY AUTO_INCREMENT,  
    CustomerType VARCHAR(50),  
    TimeOfDay VARCHAR(30),  
    CompetitorPrice DECIMAL(10,2)  
);
```

Step 3: Create Price Arms Table

Each price acts as an arm of the bandit.

```
CREATE TABLE PriceArm (  
    PriceID INT PRIMARY KEY AUTO_INCREMENT,  
    Price DECIMAL(10,2),  
    Alpha INT DEFAULT 1,  
    Beta INT DEFAULT 1  
);
```

Step 4: Create Pricing History Table

```
CREATE TABLE PricingHistory (  
    HistoryID INT PRIMARY KEY AUTO_INCREMENT,  
    ContextID INT,  
    PriceID INT,  
    Conversion BOOLEAN,  
    SaleDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY(ContextID) REFERENCES Context(ContextID),  
    FOREIGN KEY(PriceID) REFERENCES PriceArm(PriceID)  
);
```

Step 5: Insert Sample Prices

```
INSERT INTO PriceArm(Price)  
VALUES  
(90),  
(95),  
(100),  
(105),  
(110);
```

Step 6: Insert Sample Context

```
INSERT INTO Context(CustomerType,TimeOfDay,CompetitorPrice)  
VALUES  
('New Customer','Morning',95),  
('Returning Customer','Evening',100),  
('Premium Customer','Night',110);
```

Step 7: Record Customer Conversions

```
INSERT INTO PricingHistory(ContextID,PriceID,Conversion)  
VALUES  
(1,2,1),
```

(1,3,0),

(2,4,1),

(3,5,1),

(2,3,0);

Step 8: Update Beta Distribution

Success

UPDATE PriceArm

SET Alpha = Alpha + 1

WHERE PriceID = 2;

Failure

UPDATE PriceArm

SET Beta = Beta + 1

WHERE PriceID = 3;

Step 9: View Current Posterior Parameters

SELECT

PriceID,

Price,

Alpha,

Beta

FROM PriceArm;

Step 10: Category of Thompson Sampling

Posterior Mean

[
 $\text{Posterior Mean} = \frac{\alpha}{\alpha + \beta}$
]

SQL

SELECT

PriceID,

Price,
Alpha,
Beta,
ROUND(Alpha/(Alpha+Beta),3) AS ExpectedConversion
FROM PriceArm
ORDER BY ExpectedConversion DESC;

Step 11: Simulate Competitive Shock

Competitor lowers price.

```
UPDATE Context
SET CompetitorPrice=85
WHERE ContextID=1;

Adjust product price.

UPDATE PriceArm
SET Price=88
WHERE PriceID=1;
```

Step 12: Generate Pricing Report

```
SELECT
c.CustomerType,
c.TimeOfDay,
p.Price,
ph.Conversion,
p.Alpha,
p.Beta
FROM PricingHistory ph
JOIN Context c
ON ph.ContextID=c.ContextID
JOIN PriceArm p
ON ph.PriceID=p.PriceID;
```

ER Diagram

Context

ContextID (PK)

CustomerType

TimeOfDay

CompetitorPrice

|

|

| 1

|

| M

PricingHistory

HistoryID (PK)

ContextID (FK)

PriceID (FK)

Conversion

SaleDate

|

|

| M

|

| 1

PriceArm

PriceID (PK)

Price

Alpha

Beta

Features Implemented

- ✓ Designed **Context**, **PriceArm**, and **PricingHistory** tables.
- ✓ Defined **Primary Key** and **Foreign Key** relationships.
- ✓ Represented different prices as **bandit arms**.
- ✓ Stored contextual information (customer type, time, competitor price).
- ✓ Recorded customer conversions.
- ✓ Maintained **Beta distribution parameters (Alpha and Beta)**.
- ✓ Calculated expected conversion probability for each price.
- ✓ Simulated **competitive price shocks** and updated prices dynamically.
- ✓ Generated reports showing pricing decisions and conversion history.

This implementation matches the project requirements for a **Dynamic Contextual Pricing Engine using Thompson Sampling** with appropriate database design, relationships, sample data, updates, and reporting.